

Tutorial 6: Rotational Dynamics, Moment of Inertia and Angular Momentum

Rotational Dynamics, Moment of Inertia, Energy in rotation

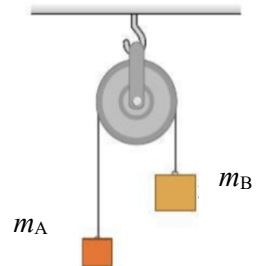
Q1.

Two solid spheres simultaneously start rolling from rest, from the same starting line, down an incline. One sphere has twice the radius and eight times the mass of the other. Which reaches the bottom of the incline first? Which has the greater speed there? Which has the greater total kinetic energy at the bottom?

Q2. (Follow-up from lecture example)

An *Atwood's machine* consists of two masses m_A and m_B , which are connected by a massless inelastic cord that passes over a pulley. The pulley has mass M and radius R and moment of inertia I about its axle.

- (Lecture example) Determine the acceleration of the masses m_A and m_B and compare your answer to the situation in which the moment of inertia of the pulley is ignored (i.e. massless pulley).
- (Follow-up) For this part, take $I = \frac{1}{2} M R^2$. What force does the ceiling exert on the hook? Is your answer greater, less than, or equal to the combined weight of the masses and pulley? Explain.



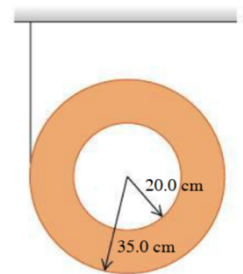
Q3.

- An automobile engine develops a torque of 255 Nm at 3750 rpm. What is the power of the engine?
- A centrifuge rotor has a moment of inertia of 42.5 g m². How much energy is required to bring it from rest to 9750 rpm?

Q4.

A thin, light string is wrapped around the outer rim of a uniform hollow cylinder of mass 4.80 kg having inner and outer radii as shown. The cylinder is then released from rest. The string does not slip on the cylinder.

- How far must the cylinder fall before its center is moving at 6.44 m/s?
- If you just dropped this cylinder without any string, how fast would its center be moving when it had fallen the distance in part (a)?
- Why do you get two different answers when the cylinder falls the same distance in both cases?



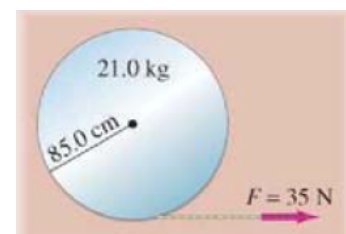
Q5.

A solid uniform disk of mass 21.0 kg and radius 85.0 cm is at rest *flat* on a frictionless surface as shown. A string is wrapped around the rim of the disk and a constant force of 35.0 N is applied to the string. The string does not slip on the rim.

- In what direction does the CM move?

When the disk has moved a distance of 5.5 m, determine

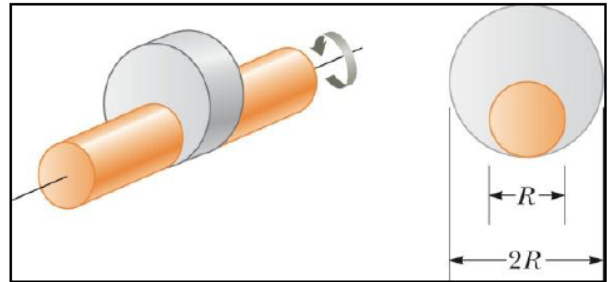
- how fast it is moving,
- how fast it is spinning (in rad/s), and
- the length of string that has unwrapped from around the rim.



Top view

Q6.

Many machines employ cams for various purposes, such as opening and closing valves. In the figure, the cam is a circular disk of radius R with a hole of diameter R cut through it. The hole does not pass through the center of the disk. The cam with the hole cut out has mass M . The cam is mounted on a uniform, solid, cylindrical shaft of diameter R and also of mass M . What is the kinetic energy of the cam-shaft combination when it is rotating with angular speed ω about the shaft's axis?



Angular Momentum

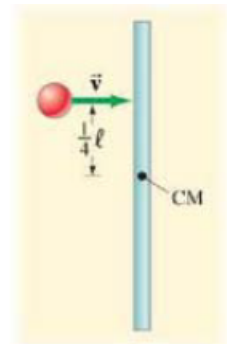
Q7.

Four identical particles of mass m are mounted at equal intervals on a thin rod of length l and mass M , with one mass at each end of the rod. If the system is rotated with angular velocity ω about an axis perpendicular to the rod through one of the end masses, determine

- the kinetic energy, and
- the angular momentum of the system.

Q8.

A thin rod of mass M and length l rests on a frictionless table and is struck at a point $l/4$ from its CM by a clay ball of mass m moving at speed v . The ball sticks to the rod. Determine the translational and rotational motion of the rod after collision.



Gyroscopic precession

Q9.

The Hubble Space Telescope is stabilized to within an angle of about 2-millionths of a degree by means of a series of gyroscopes that spin at 19,200 rpm. Although the structure of these gyroscopes is actually quite complex, we can model each of the gyroscopes as a thin-walled cylinder of mass 2.0 kg and diameter 5.0 cm, spinning about its central axis. How large a torque will it take to cause these gyroscopes to precess through an angle of 1.20×10^{-6} degree during a 5.50 hour exposure of a galaxy? Assume that the spin axis is tilted at 90 degrees to the precession axis as shown below.

Answers

- Q2. (i) $\frac{(m_b - m_a)}{(m_a + m_b + I/r^2)} g$ (ii) $\frac{g(M^2 + 3M(m_A + m_B) + 8m_A m_B)}{M + 2(m_A + m_B)}$
- Q3. 99.96 kW, 22.2 kJ
- Q4. (a) 3.52 m. (b) 8.30 m/s
- Q5. 4.3 m/s, 10 rad/s, 11 m
- Q6. $13MR^2\omega^2/24$
- Q7. $(\frac{7}{9}m + \frac{1}{6}M)\ell^2\omega^2, (\frac{14}{9}m + \frac{1}{3}M)\ell^2\omega$
- Q8. $v_{CM, final} = \frac{mv}{m + M}, \omega_{final} = \frac{12mv}{\ell(7m + 4M)}$
- Q9. $2.66 \times 10^{-12} \text{ N m}$

